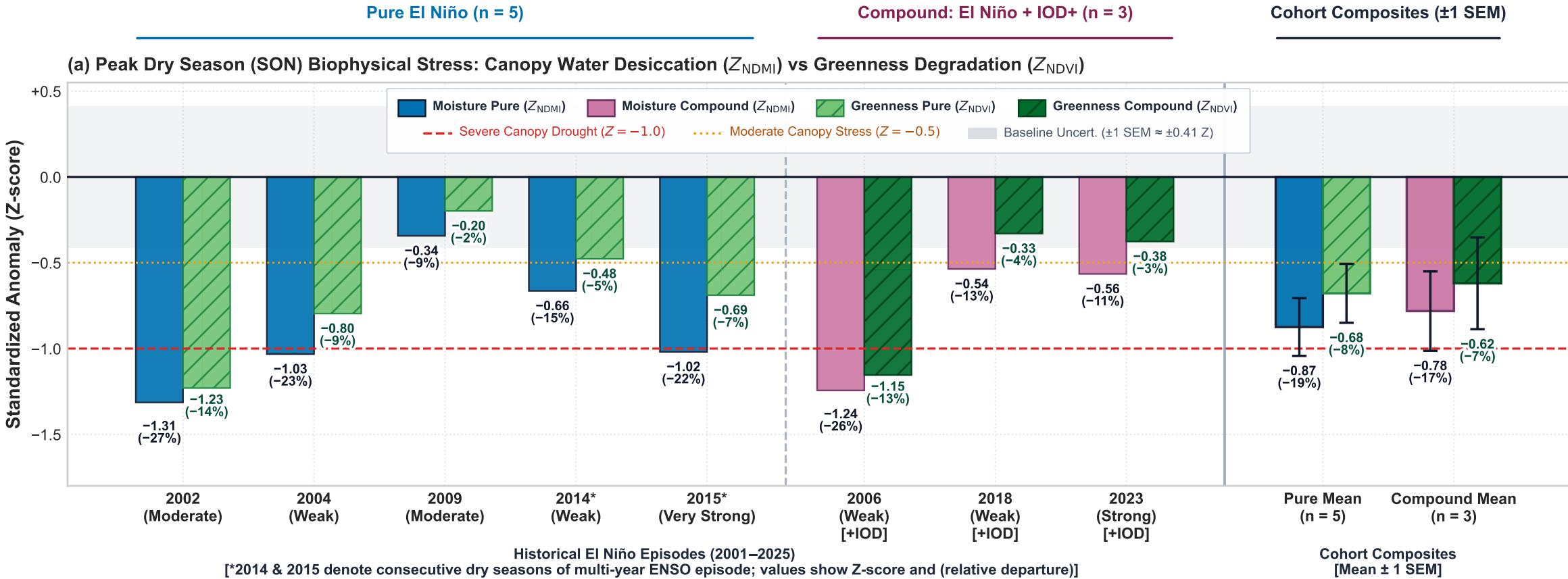
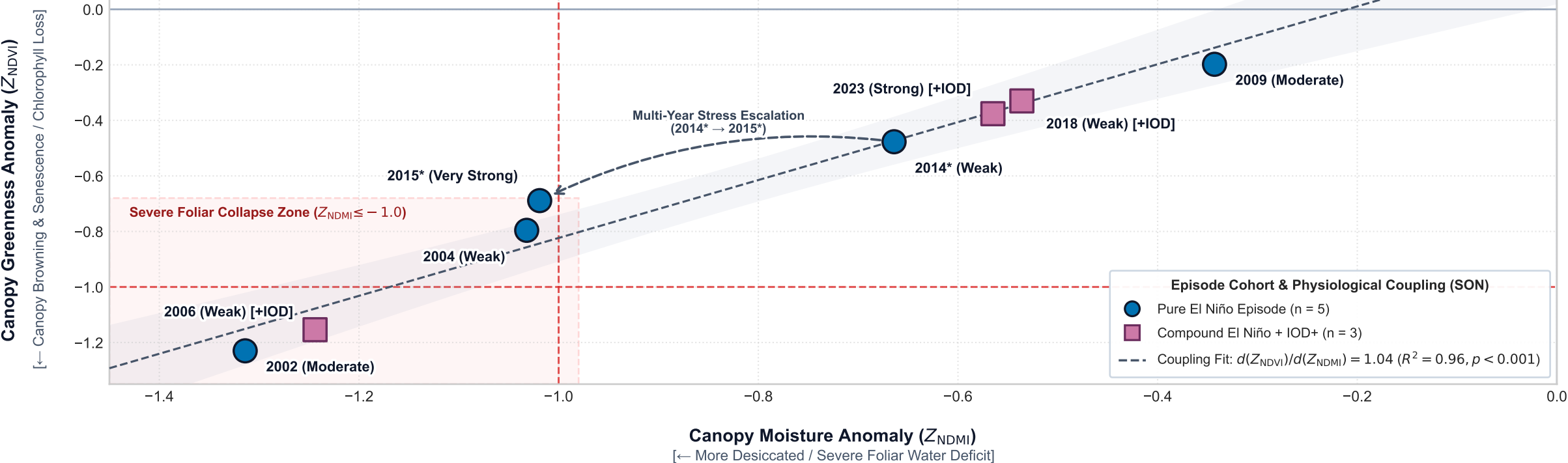


East Java Terrestrial Vegetation Response Across Historical El Niño Episodes (2001–2025)

ENSO-Neutral Baseline (n = 6; 2019 IOD+ unexcluded introduces slight conservative bias in Z-scores): NDVI = 0.545, NDMI = 0.382, EVI = 0.342 | Data: MODIS Terra MOD13A2 (16-day, 1 km)



(b) Plant Physiological Coupling: Foliar Water Deficit (Z_{NDMI}) vs Canopy Greenness Loss (Z_{NDVI}) across Historical Episodes (SON)



• Data Sources: MODIS Terra MOD13A2 16-day 1 km composites (NDVI, EVI, and NDMI liquid water band).
• ONI Tiers: Weak (+0.5–0.9°C), Mod (+1.0–1.4), Strong (+1.5–1.9), V.Strong ($\geq +2.0^\circ\text{C}$); Compound (+DMI $\geq +0.4^\circ\text{C}$).
• Episode Continuity: *2014 & 2015 = consecutive dry seasons of one ENSO episode ($n_{\text{eff}} < 8$).

• Neutral Baseline: 6 neutral years (2019 IOD+ unexcluded; baseline SEM = ± 0.41 Z).
• Plant Ecophysiology: Foliar dehydration (NDMI, 2.1 μm) leads chlorophyll breakdown & browning (NDVI).
• Statistical Rigor: Welch t (n=5 vs 3): NDMI t = -0.39 ($p = 0.72$, n.s.); NDVI t = -0.26 ($p = 0.81$, n.s.).